



Mole Concept : RACE-01

Isotopes; Isobars; Isotones; Isodiaphers; Isosters; Isoelectronic (Calculation of Electron, Proton and Neutron)

1. Species which are isoelectronic to one another are

(a) $\text{CN}^\ominus$ (b) $\text{OH}^\ominus$ (c) $\text{CH}_3^\ominus$ (d) N_2 (e) CO

Correct answer is

(A) a, b, c (B) a, c, d (C) a, d, e (D) b, c, d

CBM080

Mole Concept (Basic Introduction to Mole; Mole Calculations)

2. The actual weight of a molecule of water is -

(A) 18 g (B) 2.99×10^{-23} g
(C) both (A) & (B) are correct (D) None of these

CBM093

3. The number of atoms in 4.25 g of NH_3 is approx :-

(A) 1×10^{23} (B) 1.5×10^{23} (C) 2×10^{23} (D) 6×10^{23}

CBM095

4. Number of neutrons present in 1.7 g of ammonia is -

(A) N_A (B) $N_A/10 \times 4$ (C) $(N_A/10) \times 7$ (D) $N_A \times 10 \times 7$

CBM097

5. Sum of number of protons, electrons and neutrons in 12g of $^{12}_6\text{C}$ is :-

(A) 1.8 (B) 12.044×10^{23} (C) 1.084×10^{25} (D) 10.84×10^{24}

CBM099

6. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture?

(A) 4 : 1 (B) 16 : 1 (C) 2 : 1 (D) 1 : 4

CBM109

Ideal Gas Equation: Relative Density; Vapour Density (Dulong Petits Law)

7. At NTP (1 atm & 273 K), 5.6 lit of a gas weight 8 g. The vapour density of gas is :-

CBM114

Percentage Composition (Mass Percentage, Mole Percentage); Average Atomic Mass; Average Molecular Mass

8. A sample of oxygen gas (only O_2 molecules) contains O^{16} and O^{17} oxygen atoms only in 9 : 1 atomic ratio. The average molecular mass of O_2 , this sample, is :

(A) 32.00 (B) 16.30 (C) 32.20 (D) 33.80

CBM289

9. 20 gm pyrolusite ore exactly requires 29.2 gm HCl for complete reaction:
 $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$
 The mass percent of pure MnO_2 in the pyrolusite ore is (Mn = 55)
 (A) 17.4 % (B) 87 % (C) 43.5 % (D) 14.6 %

CBM291

10. An element has three isotopes X^{20} , X^{21} and X^{22} . Percentage abundance (mole %) of X^{20} is 90 and average atomic mass of X is 20.18. Select correct option(s) -
 (A) Percent abundance of $X^{21} = 8\%$ (B) Percent abundance of $X^{21} = 2\%$
 (C) Percent abundance of $X^{22} = 8\%$ (D) Percent abundance of $X^{22} = 2\%$

CBM292

Empirical and Molecular Formulae

11. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide is :-
 (A) M_2O (B) M_2O_3 (C) MO (D) M_3O_4

CBM105

ANSWER KEY

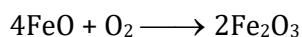
Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	D	C	D	A	16	C	B	BC
Question	11									
Answer	C									



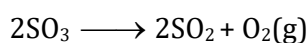
Mole Concept : RACE-02

Stoichiometry With Limiting Reagent and Excess Reagents (Combustion Based Questions; Carbon Percentage; Hydrogen Percentage)

1. Ferric oxide can be obtained by oxidation of FeO:



The O_2 gas required can be prepared by the following reaction.



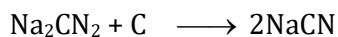
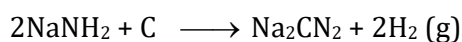
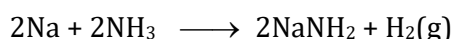
What is the maximum amount of Fe_2O_3 can be produced by 144 g FeO and 160 g SO_3 .

[Atomic mass of Fe = 56]

- (A) 0.320 g (B) 80 g (C) 120 g (D) 160 g

CBM293

2. Consider the following reaction

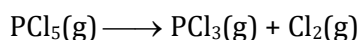


51gm dry ammonia gas is passed over excess heated sodium to form sodamide (NaNH_2) which is further reacted with carbon (excess) to finally form NaCN. Find the total volume of $\text{H}_2(\text{g})$ evolved at 0.5 atm, 273K

- (A) 201.6 L (B) 100.8 L (C) 403.2 L (D) 50.4 L

CBM294

3. $\text{PCl}_5(\text{g})$ decomposes according to following reaction,



Calculate density (in gm / l) of mixture at 24 atm & 273 K, when $\text{PCl}_5(\text{g})$ undergo 50% decomposition. [$R = 0.08 \text{ atm l mol}^{-1} \text{ K}^{-1}$]

CBM295

4. 32.2 gm of an organic compound containing C, H & O when completely combusted produces 61.6 gm of CO_2 and 37.8 gm of H_2O . Based on this select the correct option -

- (A) % of C in the organic compound is 40%
(B) % of H in the organic compound is 13.04 %
(C) % of O in the organic compound is 17.04 %
(D) Data given is impossible

CBM296

5. Match the column-I and Column-II

	Column-I (Reactions with 2 moles A and 5 moles B)		Column-II (Characteristics)
(A)	$A + 2B \rightarrow \text{products}$	(P)	'A' is limiting reagent
(B)	$A + 3B \rightarrow \text{products}$	(Q)	'B' is limiting reagent
(C)	$2A + 5B \rightarrow \text{products}$	(R)	During chemical reaction, mole ratio of reactants will not change
(D)	$A + B \rightarrow \text{products}$	(S)	3 moles of excess reactant is left after complete reaction
		(T)	1 moles of excess reactant is left after complete reaction

CBM297

Principle of Atom Conservation (POAC)

6. When 6.0 gm of a hydrocarbon (M.M. = 30) is burnt completely 17.6 gm CO_2 is formed. The molecular formula of hydrocarbon

- (A) C_2H_2 (B) C_2H_4 (C) C_2H_6 (D) $\text{C}_2\text{H}_6\text{O}$

CBM301

7. 8 gm sulphur is burnt to form SO_2 which is further oxidised by chlorine water. The solution is then treated with BaCl_2 solution. The moles of BaSO_4 precipitated is :

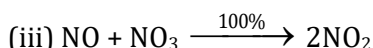
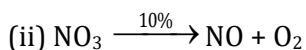
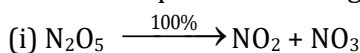
[At.wt. of S = 32]

- (A) 1 mole (B) 0.5 mole (C) 0.25 mole (D) 2 mole

CBM302

Percentage Yield; Percentage Purity

8. N_2O_5 decomposed according to the following sequence of reactions.



Calculate the number of moles of NO_2 produced due to complete decomposition of 6 mole of N_2O_5 .

Fill your answer by multiplying it with 100.

CBM304

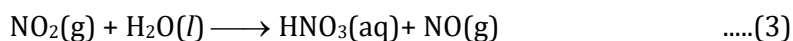
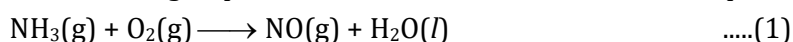
9. Match the column :

	Column-I (Reaction)		Column-II (At the end)
(A)	$2A + 2B \xrightarrow{50\% \text{ yield}} 3C$ 4 mol 6 mol	(P)	3 moles C formed
(B)	$\frac{1}{2}A + 2B \xrightarrow{80\% \text{ yield}} C$ 4 mol 8 mol	(Q)	3.2 moles C formed
(C)	$3A + 2B \xrightarrow{60\% \text{ yield}} C$ 15 mol 20 mol	(R)	A is limiting reagent
(D)	$A + 3B \xrightarrow{20\% \text{ yield}} 2C$ 5 mol 12 mol	(S)	B is limiting reagent
		(T)	1.6 moles C formed

CBM305

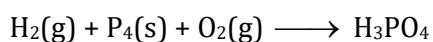
Paragraph for Q.10

The following sequence of reaction occurs in commercial production of aqueous nitric acid.



In an industry 378 kg of HNO_3 was required to be produced

10. 5.00 moles of hydrogen gas, 3 moles of white phosphorus $\{\text{P}_4(\text{s})\}$ and 12 moles of oxygen gas are taken in a sealed flask and allowed to react as follows :



Determine the moles of ortho-phosphoric acid that can be produced, considering that the reaction occurs in 90% yield.

CBM309

ANSWER KEY

Question	1	2	3	4	5	6
Answer	D	A	152.747	B	(A) $\rightarrow$ P,T; (B) $\rightarrow$ Q; (C) $\rightarrow$ R; (D) $\rightarrow$ P,S	C
Question	7	8	9		10	
Answer	C	720	(A) $\rightarrow$ (P,R) ; (B) $\rightarrow$ (Q,S); (C) $\rightarrow$ (P,R); (D) $\rightarrow$ (S,T)		3.00	



Mole Concept RACE-03

Problems based on Mixture, Sample; Impurity, and Heating Effect using Mole Concept

1. When a solid mixture of NaHCO_3 and Na_2CO_3 is strongly heated to 500K in an open vessel, the mass of sample decreases. The decrease is -
 (A) Only due to $\text{CO}_2(\text{g})$ formed by decomposition of Na_2CO_3
 (B) Only due to $\text{CO}_2(\text{g})$ formed by decomposition of NaHCO_3
 (C) Only due to $\text{CO}_2(\text{g})$ formed by decomposition of both, Na_2CO_3 and NaHCO_3
 (D) Due to $\text{CO}_2(\text{g})$ and water vapour formed by decomposition of NaHCO_3

CBM310

2. A 1800gm mixture of anhydrous $\text{CuSO}_4(\text{s})$ and its hydrated form $[\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})]$ undergoes 20% loss in mass in heating. Mole fraction of CuSO_4 in mixture (Atomic mass of Cu = 64)
 (A) $\frac{3}{40}$ (B) $\frac{4}{9}$ (C) $\frac{5}{9}$ (D) $\frac{1}{2}$

CBM311

3. 12.46 g of a mixture of MgO and MgCO_3 on strong heating lost 4.4 g in weight what is the % MgO in the mixture

CBM312

4. 1.19 gm mixture of Cu, Zn and Al contain 50% copper by weight and 3×10^{21} atoms of zinc, find gram atoms of aluminium in given mixture?
 (At. mass of Al = 27 ; Cu = 63.55 , Zn=65.0) [$N_A = 6 \times 10^{23}$]

CBM313

5. 62.5 gm of a mixture of CaCO_3 and SiO_2 are treated with excess of HCl and 1.1 gm of CO_2 is produced. What is mass % CaCO_3 in the mixture.

CBM314

Parallel Reaction; Sequential Reaction

6. Moles of ABC_3 produced in the following set of reaction when 180 gm of A, 180gm of B and 200 gm of C are mixed in a container (given molar masses of A,B,C are 20, 30 & 10 respectively).
 $2\text{A} + 3\text{B} + 5\text{C} \longrightarrow \text{A}_2\text{BC}_2 + \text{B}_2\text{C}_3$
 $\text{B}_2\text{C}_3 + 3\text{C} \longrightarrow 2\text{BC}_3$
 $\text{BC}_3 + \text{A} \longrightarrow \text{ABC}_3$
 (A) 5 (B) 4 (C) 10/3 (D) 20/3

CBM315

Introduction; Gay Lussac's Law of Combining Volumes; Eudiometry

7. 10 mL of gaseous hydrocarbon on combustion gives 40 mL of $\text{CO}_2(\text{g})$ and 50 mL of $\text{H}_2\text{O}(\text{vapour})$. The hydrocarbon is
 (A) C_4H_5 (B) C_8H_{10} (C) C_4H_8 (D) C_4H_{10}

CBM111

Molarity

8. Molarity and molality of pure CH_3COOH are respectively. ($d_{\text{CH}_3\text{COOH}} = 1.5 \text{ g/ml}$)
 (A) 16.67, 25 (B) 25, 16.67 (C) 50, 33.3 (D) 25 both

CBM317

Mixing and Dilution

9. 500 mL, 0.2M Na_2SO_4 solution is mixed with 100 mL, 17.1% $\left(\frac{w}{v}\right)$ $\text{Al}_2(\text{SO}_4)_3$ solution and resulting solution is diluted to 5 times. Find the molarity of SO_4^{2-} ions in the final solution
 [Mw of Al = 27, S = 32, Na = 23]

- (A) $\frac{1}{12}$ M (B) $\frac{5}{12}$ M (C) $\frac{7}{5}$ M (D) $\frac{12}{5}$ M

CBM318

10. 5 ml of 1.2 M- NaOH solution is diluted till the concentration becomes 2% (w/v). Assuming volume to be additive, the volume of water added for dilution is -

- (A) 12 ml (B) 7 ml (C) 19 ml (D) 5 ml

CBM320

11. 300 gm, 30% (w/w) NaOH solution is mixed with 500 gm 40% (w/w) NaOH solution. What is % (w/v) NaOH if density of final solution is 2 gm/mL :

- (A) 72.5 (B) 65 (C) 62.5 (D) None of these

CBM321

12. What volume (in ml) of 0.8 M - $\text{Ca}(\text{NO}_3)_2$ solution should be mixed with 40 ml of 0.6M - $\text{Al}(\text{NO}_3)_3$ solution to get a solution having NO_3^- ion concentration equal to 1.70 M ?

CBM322

Volume Strength of H_2O_2

13. Which of the following is incorrect for 17g/L of H_2O_2 solution?

- (A) Volume strength is 5.675
 (B) Molarity of solution is 0.5M
 (C) 1 ml of this solution gives 2.8 ml O_2 at 273K and 2 atm
 (D) The molarity of solution is 2M

CBM323

14. The volume strength of an aqueous H_2O_2 solution is '4.54V'. The concentration of solution is -

- (A) 0.2 M (B) 13.6 % (w/v)
 (C) 1.36 % (w/v) (D) 4.54 % (w/v)

CBM324

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	32.58	0.01	4.00	B	D	B	A	B
Question	11	12	13	14						
Answer	A	40	D	C						